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Lending Relationships and Loan Contract Terms

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1 Big picture

- **Research question.** Do repeated bank-borrower lending relationships improve the price and nonprice terms of syndicated loan contracts for publicly traded firms?
- **Main idea.** A lender that has dealt with the borrower before accumulates borrower-specific information that is costly, proprietary, and reusable. This information can lower adverse selection, reduce monitoring costs, and make the lender's future monitoring commitment more credible.
- **Empirical object.** The paper compares loans obtained from relationship lenders with loans obtained from nonrelationship lenders. A loan is a relationship loan when at least one lead lender on the current facility was also a lead lender to the borrower during the previous five years.
- **Main finding.** Relationship loans carry lower spreads. Across the main specifications, repeated borrowing from the same lender is associated with roughly a 10–17 basis point reduction in all-in-drawn loan spreads; some IV estimates are larger.
- **Who benefits most.** Relationship benefits are strongest for informationally opaque borrowers. They dissipate for very transparent borrowers: large firms, firms with rated public debt, and firms in the S&P 500.
- **Beyond spreads.** Relationship loans are less likely to require collateral, have maturity patterns consistent with monitoring theories, and are associated with larger loan availability.
- **Syndicated-loan insight.** Relationships matter not only because they reduce information asymmetry between borrower and lender. They also help mitigate *syndicate moral hazard*: the lead bank's incentive to shirk on monitoring when monitoring benefits are shared with other lenders.
- **Economic takeaway.** Even for public firms with access to equity markets and often public debt markets, relationship banking still creates economically meaningful contract benefits.

2 Incremental contribution of the paper

- **Extends relationship lending beyond small private firms.** Much of the earlier literature studies small private borrowers. This paper shows that relationship lending also matters for publicly listed firms that have broader financing choices.
- **Uses the syndicated loan market.** Syndicated lending creates a second information friction among lenders. This lets the paper study both borrower-lender information asymmetry and lead-lender moral hazard within syndicates.
- **Estimates the boundary between relationship and transactional lending.** The paper identifies borrower characteristics above which relationship spreads become statistically indistinguishable from nonrelationship spreads.
- **Jointly studies price and nonprice contract terms.** The paper examines spreads, collateral, maturity, and loan size, and later accounts for their simultaneous determination.
- **Uses multiple endogeneity approaches.** The paper combines propensity score matching, instrumental variables, and treatment-effects models to address endogenous relationship formation.

- **Separates relationship benefits from lock-in.** If banks captured opaque borrowers, relationship loans could be more expensive for opaque firms. Instead, benefits are larger for opaque firms, supporting the relationship-benefit view.
- **Connects relationships to monitoring commitments.** The tests using syndicate size, lead-bank allocation, and loan Herfindahl index show that relationships are especially valuable when delegated monitoring problems are severe.

3 Data and measurement

3.1 Loan data and sample

The loan-level data come from the Loan Pricing Corporation’s Dealscan database. The sample covers U.S. publicly listed nonfinancial borrowers from 1986 to 2003. Borrower accounting data are matched from CRSP and Compustat, and firms in financial services industries are excluded.

- The final classified loan sample contains **21,632** loan facilities in Table 1: **6,800** nonrelationship loans and **14,832** relationship loans.
- Loan terms include spreads, fees, facility size, maturity, collateral status, loan purpose, loan type, and syndicate structure.
- Borrower characteristics include assets, leverage, profitability, tangibility, interest coverage, current ratio, market-to-book, ratings, and S&P 500 status.
- The paper carefully tracks bank mergers and acquisitions so that lending relationships are not mechanically broken when a prior lender is acquired.

3.2 Lead lender identification

The relationship measure depends on identifying lead banks. The paper uses LPC’s lead-arranger credit field and also treats banks in major lead roles such as agent, administrative agent, arranger, and lead bank as lead lenders. Sole-lender loans automatically have a clear lead lender.

Interpretation. The lead lender is the lender most likely to screen, monitor, negotiate, and build proprietary information about the borrower. Therefore, relationships are defined at the lead-bank level rather than at the full syndicate level.

3.3 Relationship measures

For each current facility, the paper looks back five years in the borrower’s Dealscan history and constructs three measures of relationship strength.

- **REL(Dummy).** Equals one if at least one current lead lender was a lead lender on a prior loan to the same borrower during the previous five years.
- **REL(Amount).** For lead bank m and borrower i ,

$$REL(Amount)_m = \frac{\text{amount of loans by bank } m \text{ to borrower } i \text{ in prior five years}}{\text{total amount of loans by borrower } i \text{ in prior five years}}.$$

- **REL(Number)**. For lead bank m and borrower i ,

$$REL(Numbe)r)_m = \frac{\text{number of loans by bank } m \text{ to borrower } i \text{ in prior five years}}{\text{total number of loans by borrower } i \text{ in prior five years}}.$$

- If there are multiple lead banks, the paper uses the highest relationship value across the current lead banks.

3.4 Price and nonprice loan terms

- **AISD**. All-in-spread drawn, defined as the coupon spread over LIBOR on drawn amounts plus the annual fee. This is the main loan-price variable.
- **AISU, upfront fee, and annual fee**. Additional price components used in univariate comparisons.
- **Collateral**. Indicator for whether the facility is secured.
- **Maturity**. Months from facility activation to maturity.
- **Loan amount**. Facility size, also scaled by borrower assets or long-term debt to proxy for credit availability.

3.5 Information opacity variables

The paper uses several borrower transparency proxies to test whether relationship benefits are larger when outside lenders face greater information problems.

- borrower size, measured by log assets;
- whether the borrower lacks a public debt rating;
- whether the borrower is included in the S&P 500 index;
- the number of analysts following the borrower;
- a market microstructure measure of information asymmetry;
- discretionary accruals.

Interpretation. If relationships reduce information asymmetry, the relationship discount should be larger for smaller, unrated, less followed, and more opaque firms.

3.6 Syndicate moral hazard variables

The paper measures syndicate moral hazard using syndicate structure. The logic follows Sufi (2007): high monitoring needs are reflected in concentrated syndicates and larger lead-bank exposure.

- **Syndicate size**. Larger syndicates are interpreted as lower syndicate moral hazard.
- **Lead-bank allocation**. A higher lead-bank share indicates greater monitoring concerns.
- **Loan HHI**. A higher Herfindahl index of loan shares indicates a more concentrated syndicate.

4 Methodology and empirical specifications

4.1 Core hypotheses

The paper organizes the empirical tests around seven hypotheses.

- **H1.** Relationship loans carry lower spreads than nonrelationship loans.
- **H2.** Relationship benefits are larger when borrower information asymmetry is higher.
- **H3.** Above a sufficiently high borrower-quality threshold, relationship benefits become insignificant.
- **H4.** Relationship loans have lower spreads when syndicate moral hazard is high.
- **H5.** Relationship loans are less likely to require collateral.
- **H6.** The maturity effect of relationships depends on borrower quality. For low-quality borrowers, relationship banks can commit to monitoring through shorter maturity.
- **H7.** Relationship borrowers obtain larger loan amounts than comparable nonrelationship borrowers.

4.2 Baseline spread regression

The main spread regression is:

$$AISD = \beta_0 + \beta_1 REL(M) + \sum_i \beta_i LoanCharacteristics_i + \sum_j \beta_j BorrowerCharacteristics_j + \sum_k \beta_k Control_k + \varepsilon.$$

Controls include loan size, maturity, collateral, borrower size, leverage, coverage, profitability, tangibility, current ratio, market-to-book, borrower rating, loan purpose, loan type, industry, and year. In one specification, year effects are replaced by the prevailing default spread.

Interpretation. A negative coefficient on $REL(M)$ means that relationship lending is associated with lower borrowing costs after observable loan and borrower risk are controlled for.

4.3 Endogeneity of relationship formation

The paper addresses endogenous relationship formation in three ways.

- **Propensity score matching.** A probit model predicts the probability that a loan is a relationship loan using loan and borrower observables. Relationship loans are matched to nonrelationship loans with similar propensity scores.
- **Instrumental variables.** Geographic distance between borrower headquarters and lead-lender headquarters instruments for relationship formation. The idea is that shorter distance lowers information collection and monitoring costs, increasing the likelihood of a relationship.
- **Treatment-effects model.** The paper jointly models the binary relationship decision and the continuous loan-spread outcome, allowing the unobservables in the two equations to be correlated.

4.4 Opacity and boundary tests

To test H2, the paper estimates spread regressions with relationship interacted with opacity proxies:

$$AISD = \beta_0 + \beta_1 REL(M) + \beta_2 Opacity + \beta_3 REL(M) \times Opacity + \dots$$

To test H3, the paper uses the size interaction and plots the implied relationship benefit by borrower asset decile. The key object is whether the relationship coefficient becomes statistically indistinguishable from zero for high-quality and transparent borrowers.

4.5 Syndicate moral hazard tests

To test H4, the paper interacts relationship lending with syndicate moral hazard measures:

$$AISD = \beta_0 + \beta_1 REL(M) + \beta_2 MonitoringIncentive + \beta_3 REL(M) \times MonitoringIncentive + \dots$$

If relationships are credible monitoring commitments, the relationship benefit should be stronger when lead-bank monitoring incentives are more important.

4.6 Collateral regression

To test H5, the paper estimates a collateral logit:

$$\begin{aligned} Collateral = & \beta_0 + \beta_1 REL(M) + \beta_2 \log(LoanAmount) + \beta_3 Leverage \\ & + \beta_4 Tangibility + \beta_5 MarketToBook + \beta_6 LoanConcentration \\ & + \beta_7 \log(Maturity) + \dots \end{aligned}$$

A negative relationship coefficient means that relationship loans are less likely to be secured.

4.7 Maturity regression

To test H6, the paper estimates:

$$\begin{aligned} \log(Maturity) = & \beta_0 + \beta_1 REL(M) + \beta_2 \log(LoanAmount) + \beta_3 Leverage \\ & + \beta_4 \log(Assets) + \beta_5 MarketToBook + \beta_6 \log(AssetMaturity) \\ & + \beta_7 Regulated + \beta_8 Collateral + \dots \end{aligned}$$

The paper then splits relationship effects by borrower quality, using asset terciles and credit-rating groups.

4.8 Joint determination of contract terms

Because loan spread, maturity, and collateral can be jointly determined, the paper also estimates a system with instrumental variables:

$$\begin{aligned} AISD &= \gamma_A REL + \gamma_{AC} Collateral + \gamma_{AM} \log(Maturity) + X_A \beta_A + \epsilon_A, \\ Collateral &= \gamma_C REL + \gamma_{CM} \log(Maturity) + X_C \beta_C + \epsilon_C, \\ \log(Maturity) &= \gamma_M REL + \gamma_{MC} Collateral + X_M \beta_M + \epsilon_M. \end{aligned}$$

Instruments include default spreads or recent average loan spreads for AISD, asset maturity and regulated-industry status for maturity, and loan concentration for collateral.

4.9 Credit availability regression

To test H7, the paper uses loan amount scaled by assets or long-term debt:

$$\frac{LoanAmount}{Assets} = \beta_0 + \beta_1 REL(M) + \sum_i \beta_i LoanCharacteristics_i + \sum_j \beta_j BorrowerCharacteristics_j + \sum_k \beta_k Control_k + \varepsilon.$$

A positive coefficient means that relationship borrowers obtain larger facilities relative to borrower size.

5 Identification and interpretation

5.1 What the paper can identify

The paper identifies whether repeated borrowing from the same lead lender is associated with better loan terms after controlling for borrower risk, loan characteristics, and several sources of relationship endogeneity. The evidence is strongest as a large-scale observational analysis supported by matching, IV, and treatment-effects approaches.

5.2 Why public firms are a demanding setting

Public firms already have access to equity markets, and many have access to public debt markets. Finding relationship benefits in this setting means that relationship banking is not only a small-private-firm phenomenon. It also matters in large corporate syndicated lending.

5.3 Relationship benefit versus lock-in

The evidence favors relationship benefits over lock-in. If lenders captured opaque borrowers, relationship loans should become more expensive or less beneficial when opacity rises. Instead, relationship benefits are larger for opaque borrowers, and relationship cost savings appear even in single-lender subsamples where lock-in concerns are most relevant.

5.4 Role of distance as an instrument

Distance is used because soft information and monitoring are easier when lenders are geographically close to borrowers. The exclusion restriction is that distance affects loan spreads only through the likelihood of forming a lending relationship. This is plausible but not mechanically testable, so the IV results should be read together with matching and treatment-effects evidence.

5.5 Interpreting the boundary between relationship and transactional lending

The boundary result is an economic classification, not a legal one. For very transparent firms, the observed spread difference between relationship and nonrelationship loans disappears. The paper calls such loans transactional because proprietary lender information no longer produces a measurable spread advantage.

5.6 Why syndicate structure matters

In syndicated loans, the lead lender monitors but does not hold the entire loan. This creates a monitoring externality. A past relationship can make the lead lender's monitoring commitment more credible because the relationship lender has lower monitoring costs and more borrower-specific information.

5.7 Limitations

The paper relies on observational loan data. Relationship formation is not randomly assigned, and distance-based IV assumptions cannot be proven. Relationship benefits could also appear in other bundled services, not only loan spreads. In addition, missing or imperfect Dealscan information on collateral and syndicate shares can limit some tests.

6 Tables and figures

6.1 Table 1: Descriptive statistics of loan facilities

Read:

- Table 1 separates relationship and nonrelationship loans by year, borrower industry, and loan purpose.
- The classified sample contains 21,632 facilities: 14,832 relationship loans and 6,800 nonrelationship loans.
- Relationship loans become more common over time, reflecting both market growth and repeated borrowing patterns.
- Corporate purposes and debt repayment are the most common stated loan purposes.

Economic message: The paper studies a large, broad sample of syndicated loans. Relationship lending is common, not exceptional, in the market for public-firm bank loans.

6.2 Table 2: Summary statistics for loan and borrower characteristics

Read:

- The median AISD is 212.5 basis points.
- The median facility size is \$50 million, and the median maturity is 36 months.

- 82% of observed facilities are secured, and 79% are syndicated.
- The median borrower has \$360.4 million in assets, and 73% of facilities involve borrowers that are not rated.

Economic message: The sample contains substantial heterogeneity in loan prices, maturity, collateral, syndication, borrower size, and transparency. This variation is central for the paper’s tests.

Table 1
Descriptive statistics of loan facilities

Panel A: Calendar time distribution of loans			
Purpose	No relationship <i>REL(Dummy)</i> = 0	Relationship <i>REL(Dummy)</i> = 1	Total
1986	1	2	3
1987	68	48	116
1988	217	211	428
1989	252	317	569
1990	242	389	631
1991	276	447	723
1992	363	563	926
1993	412	737	1,149
1994	421	924	1,345
1995	359	1,018	1,377
1996	598	1,110	1,708
1997	618	1,497	2,115
1998	609	1,366	1,975
1999	548	1,272	1,820
2000	519	1,424	1,943
2001	474	1,270	1,744
2002	448	1,135	1,583
2003	375	1,102	1,477
Total	6,800	14,832	21,632

Panel B: Industry classification of borrowers			
One-digit SIC code	No relationship <i>REL(Dummy)</i> = 0	Relationship <i>REL(Dummy)</i> = 1	Total
0	23	54	77
1	372	915	1,287
2	950	2,210	3,160
3	1,706	3,357	5,063
4	682	1,820	2,502
5	895	1,928	2,823
7	602	1,360	1,962
8	229	539	768
9	31	40	71
Total	5,490	12,223	17,713

Panel C: Loan purpose			
Purpose	No relationship <i>REL(Dummy)</i> = 0	Relationship <i>REL(Dummy)</i> = 1	Total
Acquisition line	268	646	914
Corporate purposes	1,973	3,974	5,947
CP backup	190	1,483	1,673
Debt repayment	1,599	3,906	5,505
Debtor-in-possession	98	197	295
LBO/MBO	339	304	643
Recapitalization	165	297	462
Takeover	735	1,621	2,356
Working capital	1,066	1,813	2,879
Other	366	590	956
Total	6,799	14,831	21,630

Panels A through C provide the descriptive statistics for the sample of loan facilities. The statistics are reported separately for loans from relationship lenders *REL(Dummy)* = 1 and loans from nonrelationship lenders *REL(Dummy)* = 0. For any particular loan facility, *REL(Dummy)* equals 1 if any of the lead lenders for that loan facility had been a lead lender on any loans to that borrower in the five years preceding the loan facility.

Table 2
Summary statistics for key loan and borrower characteristics

Variable	N	Mean	Std. dev.	Min.	25th percentile	Median	75th percentile	Max
Panel A: Loan characteristics								
AISD (basis points)	26,955	216.63	139.89	17.50	100.00	212.50	300.00	630.00
AISU (basis points)	15,265	33.38	23.45	4.00	15.00	30.00	50.00	135.00
Fee upfront (basis points)	8,969	66.37	72.94	2.50	20.00	50.00	100.00	315.00
Fee annual (basis points)	6,634	19.52	25.63	1.33	7.50	12.50	24.50	100.00
Loan facility amount (\$ millions)	31,384	190.0	500.0	0.5	13.0	50.0	180.0	2000.0
Maturity of loan (months)	28,335	43.2	28.36	3	17	36	60	120
Collateral	19,813	0.82	0.39	0.0	1.0	1.0	1.0	1.0
Syndicate	31,387	0.79	0.41	0.0	1.0	1.0	1.0	1.0
Panel B: Borrower characteristics								
Borrower assets (\$ millions)	23,679	2,941.4	11,279.0	6.6	93.3	360.4	1,623.5	38,750.0
Coverage	22,604	40.18	731.51	0	2.4	4.83	10.05	309.40
Leverage	23,490	0.25	0.19	0	0.08	0.23	0.37	0.82
Profitability	23,838	0.14	0.13	0	0.06	0.12	0.19	0.62
Tangibility	23,596	0.34	0.23	0.02	0.16	0.29	0.50	0.90
Current ratio	22,578	2.08	2.50	0.32	1.14	1.66	2.41	8.86
Market-to-book	23,463	1.75	1.45	0.66	1.09	1.36	1.90	7.22
Investment grade	8,590	0.50	0.50	0.0	0.0	0.0	1.0	1.0
Not rated	31,387	0.73	0.45	0.0	0.0	1.0	1.0	1.0

The table provides summary statistics of various loan and borrower characteristics. Panel A reports key characteristics of the loan facilities, and panel B reports the same for borrower characteristics. AISD is the “all-in-spread drawn,” which is the all-inclusive cost of a drawn loan to the borrower. This equals the coupon spread over LIBOR on the drawn amount plus the annual fee and is reported in basis points. Loan Facility Size is the dollar amount of loan facility in millions. AISU is the fee charged on the undrawn loan amounts. AISD, AISU, and Annual Fee are reported in basis points. Maturity is length in months between facility activation date and maturity date. Syndicate, Collateral, and Investment Grade are the percentage of facilities that have the stated attribute. To be classified as Investment Grade, the loan has to be rated BBB or above by S&P and zero otherwise. The descriptive statistics for Investment Grade are estimated for the subset of firms that are rated. Asset is the book value of assets of the borrower as reported in Compustat. Leverage is the ratio of book value of total debt to book value of total assets. Coverage is the ratio of EBITDA to interest expenses. Profitability is the ratio of EBITDA to sales. Tangibility is the ratio of NPPE to total assets. Current ratio is the ratio of current assets to current liabilities. Market-to-book is the ratio of (book value of assets – book value of equity + market value of equity) to the book value of assets. All values are winsorized at the 1% and 99% levels.

Table 1: Descriptive statistics of loan facilities.

Table 2: Summary statistics for key loan and borrower characteristics.

6.3 Tables 3 and 4: Univariate and multivariate spread evidence

Read Table 3:

- Relationship loans have much lower average AISD: 187.47 bps versus 238.97 bps for nonrelationship loans.
- Relationship loans have lower undrawn spreads, lower upfront fees, and lower annual fees.
- Relationship loans are larger and less likely to be secured.

- Relationship borrowers are larger, more profitable, more likely to be rated, and more likely to be investment grade.

Read Table 4:

- After loan and borrower controls, relationship loans still have spreads about 10–11 bps lower.
- The result holds using REL(Dummy), REL(Number), and REL(Amount).
- Collateral is associated with higher spreads, and longer maturity is associated with lower spreads in the OLS specifications.
- Adding the default spread instead of year dummies leaves the relationship coefficient strongly negative.

Economic message: Raw spread differences are large, partly because relationship borrowers differ from nonrelationship borrowers. But even after controls, relationships retain an economically meaningful pricing benefit.

Table 3
Key price and nonprice loan contract terms—relationship vs. nonrelationship loans

	No relationship <i>REL(Dummy)</i> = 0 (A)	Relationship <i>REL(Dummy)</i> = 1 (B)	<i>t</i> -statistic (A)-(B) (<i>z</i> -statistic for Wilcoxon's sum test)
Panel A: Loan characteristics			
AISD (bps)	238.97 (250.00)	187.47 (175.00)	23.86*** (23.08***)
AISU (bps)	37.36 (37.50)	30.22 (25.00)	14.56*** (17.43***)
Upfront fee (bps)	76.75 (50.00)	53.16 (30.00)	12.15*** (14.58***)
Annual fee (bps)	23.52 (12.50)	17.07 (12.50)	8.16*** (6.12***)
Facility size (\$ millions)	125.00 (40.00)	271.00 (100.00)	-18.56*** (-34.89***)
Maturity (months)	44.84 (36.00)	42.35 (36.00)	5.84*** (5.02***)
Collateral	0.84 (1.00)	0.77 (1.00)	10.43*** (10.38***)
Syndicate	0.78 (1.00)	0.90 (1.00)	-22.95*** (-22.68***)
Panel B: Borrower characteristics			
Total assets (\$ millions)	2125.67 (279.82)	4074.99 (714.03)	-10.00*** (-26.94***)
Coverage=(EBITDA/Interest)	36.64 (4.38)	33.50 (5.06)	0.27 (-10.04***)
Leverage=LT Debt/Total assets	0.25 (0.23)	0.27 (0.26)	-7.24*** (-8.59***)
Profitability=EBITDA/Sales	0.14 (0.11)	0.16 (0.13)	-9.69*** (-13.08***)
Tangibility=NPPE/Total Assets	0.35 (0.29)	0.35 (0.30)	-2.30** (-10.80***)
Current ratio=CA/CL	1.96 (1.64)	1.90 (1.58)	2.50** (17.92***)
Market-to-book	1.68 (1.31)	1.73 (1.39)	-2.15*** (-8.55***)
Investment grade	0.42 (0.00)	0.51 (1.00)	-6.22*** (-6.20**)
Not rated	0.75 (1.00)	0.62 (1.00)	19.86*** (19.68***)

Panel A segregates the entire sample into nonrelationship (*REL(Dummy)* = 0) and relationship (*REL(Dummy)* = 1) loans. The first two columns report the mean (medians in parentheses) values for various price and nonprice terms of loan contract. Panel B provides similar details for borrower-specific characteristics. The last column provides the *t*-statistic for difference in means (*z*-statistic for Wilcoxon's rank sum test). All values are winsorized at the 1% and 99% levels. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Other loan-specific features reported include loan facility size, maturity, and collateral (denoted as percentage of facilities secured). The last column reports the differences in mean (median) loan characteristics between relationship and nonrelationship loans. The results of univariate tests of differences in means and medians provide strong evidence that relationship loans enjoy significantly better loan price terms, as well as nonprice terms such as loan size and collateral requirement. Comparing AISD (the most comprehensive measure of the borrowing cost) for a company borrowing from a relationship lender, we find that

Table 3: Relationship versus nonrelationship loan terms.

Table 4
Effect of lending relationships on cost of borrowing

	(1)	(2)	(3)	(4)
Const.	903.21*** (22.76)	904.42*** (22.95)	903.82*** (22.89)	331.57*** (19.40)
<i>REL(Dummy)</i>	-10.55*** (2.44)			-11.15*** (2.47)
<i>REL(Number)</i>		-9.94*** (2.72)		
<i>REL(Amount)</i>			-10.15*** (2.68)	
Log(Maturity)	-18.07*** (2.54)	-17.91*** (2.54)	-17.95*** (2.54)	-18.58*** (2.46)
Collateral	60.01*** (2.87)	59.99*** (2.87)	59.99*** (2.87)	58.82*** (2.70)
Default spread				34.47*** (1.90)
Log(Loan size)	-16.17*** (1.56)	-16.17*** (1.55)	-16.05*** (1.55)	-16.07*** (1.23)
Log(Assets)	-9.71*** (1.57)	-10.06*** (1.56)	-10.05*** (1.56)	-9.55*** (1.16)
Log(1+Coverage)	-23.91*** (1.69)	-23.93*** (1.70)	-23.93*** (1.70)	-24.19*** (1.56)
Leverage	21.63*** (8.73)	21.21** (8.71)	21.44*** (8.72)	23.92*** (7.98)
Profitability	-44.03*** (13.65)	-43.68*** (13.64)	-44.00*** (13.62)	-41.86*** (11.29)
Tangibility	-12.00* (6.73)	-11.78* (6.76)	-11.81* (6.76)	-13.31** (5.29)
Current ratio	-2.71*** (0.74)	-2.71*** (0.74)	-2.71*** (0.74)	-2.45*** (0.71)
Market-to-book	-1.13 (0.96)	-1.12 (0.96)	-1.12 (0.96)	-1.95** (0.92)
Obs.	13,158	13,158	13,158	13,158
R ²	0.57	0.57	0.57	0.56

The table provides the OLS estimates (corrected for heteroscedasticity and clustering) of the following equation:

$$AISD = \beta_0 + \beta_1(REL(M)) + \sum \beta_i(Loan.Characteristics_i) + \sum \beta_j(Borrower.Characteristics_j) + \sum \beta_k(Controls_k).$$

The dependent variable AISD is the coupon spread over LIBOR on the drawn amount plus the annual fee in bps. *REL(M)* is the measure of relationship strength, estimated in three different ways: *REL(Dummy)* (1 if there is a relationship with any of the lead banks in the last five years before the present loan and 0 otherwise), *REL(Number)* (ratio of number of deals with the lead bank(s) to total number of loans borrowed by the firm in the last five years before the current loan), and *REL(Amount)* (ratio of the dollar value of deals with the lead bank(s) to the total dollar value of loans borrowed by the firm in the last five years before the current loan). For a facility with multiple lead banks, the maximum *REL(M)* value among all the lead banks is used. Maturity is the length in months between facility activation date and maturity date. Loan amount is the loan facility size in millions of real year 2000 dollars. Collateral is a dummy variable that equals 1 if the facility has the stated attribute and 0 otherwise. Log(Assets) is the natural log of book value of assets in real year 2000 dollars of the borrower as reported in Compustat. Leverage is the ratio of the book value of total debt to the book value of assets. Coverage is the ratio of EBITDA to interest expenses. Profitability is the ratio of EBITDA to sales. Tangibility is the ratio of NPPE to total assets. Current ratio is the ratio of current assets to current liabilities. Market-to-book is the ratio of (book value of assets - book value of equity + market value of equity) to the book value of assets. In addition to the variables reported, the regression also includes industry dummies based on the one-digit SIC code of the borrower, dummies for stated purpose of the facility, dummies for loan type, calendar year dummies, and dummies for the credit rating of the borrower, with not-rated firms considered a separate group. Numbers in parentheses are standard errors corrected for heteroscedasticity and firm-level clustering. Clustering at the deal level produces similar results. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 4: Lending relationships and cost of borrowing.

6.4 Table 5: Endogeneity of lending relationships

Read:

- Propensity score matching shows relationship loans are about 11–12 bps cheaper than matched nonrelationship loans.
- The benefit is larger when the comparison group consists of borrowers that break an existing relationship.
- The IV estimate using log distance as an instrument produces a larger relationship effect, about -56.9 bps.
- The treatment-effects model gives a relationship effect of -16.91 bps.
- Distance is negatively related to relationship formation, consistent with soft-information theories.

Economic message: The spread benefit is not eliminated when the paper addresses endogenous relationship formation. If anything, relationship benefits become larger in the IV and treatment-effects specifications.

Table 5
Endogeneity of lending relationships

Panel A: Propensity score matching estimation				
Estimator	1	2	3	
NEAREST NEIGHBOR (<i>n</i> = 10)	-12.27*** (2.93)***	-13.65*** (2.64)	-7.69 (4.77)	
NEAREST NEIGHBOR (<i>n</i> = 50)	-10.94*** (2.44)	-12.52*** (2.41)	-7.02* (4.25)	
GAUSSIAN	-11.16*** (2.37)	-12.56*** (1.84)	-8.64** (3.85)	
EPANECHNIKOV	-10.95*** (2.49)	-12.53*** (1.83)	-7.82** (3.55)	

Panel B: Instrumental variables regression estimation of effect of relationship on AISD				
First Stage Regression		Instrumental variables estimation		
		OLS	IV	
Constant	-4.552*** (0.001)	Constant	331.57*** (19.40)	422.16*** (23.80)
Log(1+distance)	-0.058*** (0.011)	<i>REL(Dummy)</i>	-11.15*** (2.47)	-56.90** (25.14)
Log(Assets)	-0.002 (0.015)	Log(Maturity)	-18.58*** (2.46)	-15.40*** (2.32)
Log(Loan size)	0.175*** (0.015)	Collateral	58.82*** (2.70)	71.41*** (3.25)
Profitability	-0.108 (0.168)	Default spread	34.47*** (1.90)	35.71*** (2.50)
Tangibility	-0.083 (0.082)	Log(Loan size)	-16.07*** (1.23)	-11.63*** (2.01)
Leverage	0.443*** (0.098)	Log(Assets)	-9.55*** (1.16)	-12.59*** (1.49)
Log(1+Coverage)	0.096*** (0.016)	Log(1+Coverage)	-24.19*** (1.56)	-24.12*** (2.12)
Current ratio	0.010 (0.019)	Leverage	23.92*** (7.98)	55.56*** (11.07)
Market-to-book	-0.007 (0.011)	Profitability	-41.86*** (11.296)	-57.85*** (15.77)
Observations	16,074	Tangibility	-13.31** (5.29)	-20.14*** (7.27)
Pseudo <i>R</i> ²	0.086	Current ratio	-2.45*** (0.71)	-1.70** (0.80)
		Market-to-book	-1.95** (0.92)	-3.23*** (1.02)
		Obs.	13,158	13,158
		<i>R</i> ²	0.56	0.49
		First-stage <i>F</i> -statistic		128.18***

Panel C: Treatment effects model		
	Relationship	AISD
Log(Assets)	0.04** (0.02)	-12.19*** (1.45)
Log(1+Coverage)	0.11*** (0.02)	-25.81*** (1.86)
Leverage	0.559*** (0.11)	45.86*** (9.67)
Profitability	-0.16 (0.19)	-57.34*** (15.53)
Tangibility	-0.15* (0.09)	-17.13** (7.17)
Current ratio	0.01 (0.01)	-1.81** (0.79)
Market-to-book	0.003 (0.01)	-3.13*** (1.03)
Log(Maturity)		-12.79*** (1.73)
Collateral		72.07*** (3.04)
Obs.	13,158	13,158
ρ	0.042	
LR test of $\rho = 0$	$\chi^2(1) = 4.84$	
Probability > χ^2	$= 0.028$	

(continued overleaf)

Table 5, part 1: Propensity-score matching and IV evidence.

(Continued)

Panel C: Treatment effects model		
	Relationship	AISD
Log(Assets)	0.04** (0.02)	-12.19*** (1.45)
Log(1+Coverage)	0.11*** (0.02)	-25.81*** (1.86)
Leverage	0.559*** (0.11)	45.86*** (9.67)
Profitability	-0.16 (0.19)	-57.34*** (15.53)
Tangibility	-0.15* (0.09)	-17.13** (7.17)
Current ratio	0.01 (0.01)	-1.81** (0.79)
Market-to-book	0.003 (0.01)	-3.13*** (1.03)
Log(Maturity)		-12.79*** (1.73)
Collateral		72.07*** (3.04)
Obs.	13,158	13,158
ρ	0.042	
LR test of $\rho = 0$	$\chi^2(1) = 4.84$	
Probability > χ^2	$= 0.028$	

Panel A reports the average difference in AISD of (i) relationship loans and nonrelationship loans; (ii) relationship loans and loans by borrowers who had lending relationships in the past but obtained the loan from a nonrelationship lender; and (iii) relationship loans and loans by borrowers who had no prior lending relationships. AISD is the coupon spread on the drawn amount plus the annual fee in bps. To examine mean AISD spread differences, we control for various borrower and lender characteristics. We compute propensity scores using the following probit model:

$$REL = \beta_0 + \sum \beta_1 (Loan_Characteristics_i) + \sum \beta_2 (Borrower_Characteristics_j) + \sum \beta_3 (Control_k).$$

The dependent variable is *REL(Dummy)*, a dummy variable that equals 1 if there is a past relationship with any of the

Table 5, part 2: Treatment-effects model.

6.5 Table 6 and Figure 1: Information opacity and the boundary of relationships

Read Table 6:

- Relationship benefits decline with borrower size: the REL(Dummy) interaction with log assets is positive and significant.
- Not-rated borrowers receive an additional relationship discount relative to rated borrowers.
- S&P 500 borrowers have a positive interaction that offsets the base relationship benefit.
- More analyst coverage reduces relationship benefits.
- Additional opacity measures, including information asymmetry and discretionary accruals, also point in the same direction.

Read Figure 1:

- Relationship benefits are economically large for smaller borrowers.
- For borrowers at the 10th asset percentile, the implied relationship discount is about 24 bps.
- The benefit becomes statistically insignificant around the top 30% of firms by asset size.
- At the 90th percentile, the implied relationship effect is close to zero.

Economic message: Relationships are most valuable when outside lenders face information problems. For very transparent borrowers, relationship lending becomes economically similar to transactional lending.

Ann. Columns 3 and 4 employ additional proxies of information opacity, a

Table 6
Borrower information opacity and benefits of relationship

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Const.	927.16*** (23.66)	902.33*** (22.75)	436.46*** (19.88)	903.75*** (23.22)	909.74*** (24.54)	518.53*** (18.49)	903.09*** (22.67)
<i>REL(Dummy)</i>	-43.77*** (8.94)	-2.35 (3.65)	-14.42*** (2.72)	-14.81*** (3.07)	-9.99*** (2.71)	-6.39* (3.50)	-14.39*** (2.74)
<i>REL(Dummy)</i> × <i>Log(Assets)</i>	5.50*** (1.37)						
<i>REL(Dummy)</i> × Not rated		-12.37*** (4.73)					
<i>REL(Dummy)</i> × S&P 500			26.94*** (6.27)				
<i>REL(Dummy)</i> × No. of analysts				0.93*** (0.30)			
<i>REL(Dummy)</i> × <i>ASY</i>					-3.23** (1.65)		
<i>REL(Dummy)</i> × Accruals						-62.60* (34.77)	
<i>REL(Dummy)</i> × Big-3 bank							22.98*** (5.28)
S&P 500		-12.26 (8.17)					
No. of analysts				-0.89*** (0.32)			
<i>ASY</i>					2.14* (1.29)		
Discretionary accruals						99.90*** (33.57)	
Big-3 bank							-20.59*** (4.64)
<i>Log(Maturity)</i>	-18.27*** (2.55)	-18.06*** (2.54)	-18.07*** (2.63)	-18.16*** (2.55)	-18.49*** (2.98)	-20.05*** (3.11)	-18.04*** (2.55)
<i>Collateral</i>	59.78*** (2.86)	59.80*** (2.86)	57.17*** (2.90)	59.82*** (2.86)	55.92*** (3.14)	57.64*** (3.46)	59.80*** (2.86)
<i>Log(Assets)</i>	-13.53*** (1.79)	-9.73*** (1.56)	-10.35*** (1.59)	-9.31*** (1.62)	-8.72*** (1.66)	-7.88*** (1.86)	-9.81*** (1.57)
<i>Log(Loan size)</i>	-16.15*** (1.55)	-16.15*** (1.55)	-16.71*** (1.60)	-16.10*** (1.55)	-17.70*** (1.63)	-17.89*** (1.82)	-15.89*** (1.55)
<i>Log(1+Coverage)</i>	-23.52*** (1.70)	-23.76*** (1.69)	-24.48*** (1.73)	-23.70*** (1.70)	-25.40*** (1.83)	-29.09*** (2.37)	-23.84*** (1.69)
<i>Leverage</i>	22.77*** (8.74)	22.04** (8.74)	18.69** (8.86)	21.27** (8.74)	12.75 (9.25)	21.19** (10.39)	21.13** (8.70)
<i>Profitability</i>	-43.81*** (13.61)	-44.26*** (13.64)	-43.08*** (14.40)	-43.49*** (13.70)	-41.30*** (15.62)	-16.89 (16.71)	-43.52*** (13.61)
<i>Tangibility</i>	-12.10* (6.69)	-11.95* (6.73)	-2.35 (6.91)	-11.93* (6.72)	-7.10 (7.11)	-6.32 (7.70)	-11.94* (6.73)
<i>Current ratio</i>	-2.65*** (0.73)	-2.72*** (0.73)	-2.73*** (0.77)	-2.68*** (0.73)	-3.13*** (0.82)	-3.09*** (1.04)	-2.71*** (0.73)

(continued overleaf)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Market-to-book	-1.23 (0.95)	-1.15 (0.96)	-1.49 (0.98)	-1.07 (0.97)	-1.19 (1.00)	-1.06 (1.36)	-1.11 (0.96)
Obs.	13,158	13,158	12,180	13,158	10,744	9,170	13,158
R ²	0.57	0.57	0.58	0.57	0.59	0.60	0.57

This table provides the OLS regression (corrected for heteroscedasticity and clustering) estimates of the following equation:

$$\begin{aligned}
 AISD = & \beta_0 + \beta_1(REL(M)) + \beta_2(Borrower\ Information\ Opacity) \\
 & + \beta_3(REL(M)) \times (Borrower\ Information\ Opacity) \\
 & + \sum \beta_i(Loan_Characteristics_i) + \sum \beta_j(Borrower_Characteristics_j) + \sum \beta_k(Controls_k).
 \end{aligned}$$

The dependent variable AISD is the coupon spread on the drawn amount plus the annual fee in bps. *REL(Dummy)* equals 1 if there is a relationship with any of the lead banks in the last five years before the present loan and 0 otherwise. Maturity is length in months between facility activation date and maturity date. Loan Amount is the loan facility size in millions of real year 2000 dollars. Collateral is a dummy variable that equals 1 if the facility has the stated attribute and 0 otherwise. Log(assets) is the natural log of book value of assets of the borrower in millions of year 2000 real dollars as reported in Compustat. Leverage is the ratio of the book value of total debt to the book value of assets. Coverage is the ratio of EBITDA to interest expenses. Profitability is the ratio of EBITDA to sales. Tangibility is the ratio of NPPE to total assets. Current ratio is the ratio of current assets to current liabilities. Market-to-book is the ratio of (book value of assets – book value of equity + market value of equity) to the book value of assets. Borrower information opacity is measured by six different proxies: Borrower's size (log of assets), if the borrower is not rated (not-rated dummy), if the borrower is in the S&P 500 index, number of analysts following the borrower firm, a microstructure measure of information asymmetry ASY developed by Bhurath, Pasquarello, and Wu (2009), and discretionary accruals computed according to Dechow and Dichev (2002). Big-3 bank is a dummy variable if the lead bank is either Citibank, JP Morgan Chase, or Bank of America. In addition to the variables reported, the regression also includes industry dummies based on the one-digit SIC code of the borrower, dummies for stated purpose of the facility, dummies for loan type, calendar year dummies, and dummies for the credit rating of the borrower, with not-rated firms considered a separate group. Numbers in parentheses are standard errors corrected for heteroscedasticity and firm-level clustering. Clustering at the deal level produces similar results. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 6, part 1: Borrower opacity and relationship benefits.

Table 6, part 2: Continued opacity results and definitions.

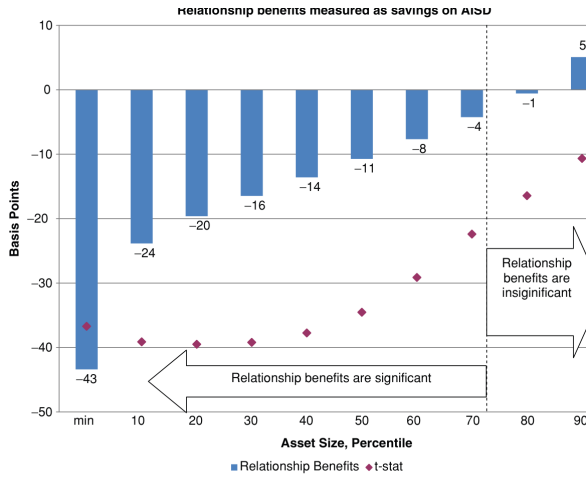


Figure 1
Relationship benefits and borrower size

Figure 1: Relationship benefits by borrower asset size.

Table 7
Role of relationships as a bank's commitment to monitor

	Panel A				
	(1)	(2)	(3)	(4)	(5)
Const.	569.22*** (60.96)	642.95*** (119.35)	898.58*** (23.18)	855.71*** (38.02)	834.47*** (43.21)
REL(Dummy)	-18.19*** (6.12)	-14.44** (7.36)	-20.27*** (3.70)	6.25 (4.96)	3.57 (4.52)
REL(Dummy) × Log(Syndicate size)			8.59*** (2.02)		
Log(Syndicate Size)			-7.63*** (2.17)		
REL(Dummy) × Lead bank allocation				-0.26*** (0.09)	
Lead bank allocation				0.49*** (0.09)	
REL(Dummy) × Loan HHI					-0.002*** (0.0008)
Loan HHI					0.005*** (0.001)
Log(Maturity)	-16.21*** (5.81)	-19.15*** (6.44)	-17.82*** (2.57)	-15.39*** (3.61)	-14.67*** (3.75)
Collateral	63.45*** (7.75)	64.70*** (8.53)	59.86*** (2.88)	60.82*** (3.62)	60.11*** (3.72)
Log(Assets)	-21.10*** (3.75)	-17.00*** (4.90)	-9.82*** (1.59)	-15.50*** (1.80)	-15.33*** (1.89)
Log(Loan size)	-20.29*** (4.24)	-25.61*** (5.04)	-15.34*** (1.63)	-11.37*** (2.00)	-10.49*** (2.21)
Log(1+Coverage)	-15.07*** (2.71)	-13.89*** (2.94)	-23.58*** (1.70)	-24.18*** (2.09)	-24.24*** (2.15)
Leverage	54.82** (22.46)	43.82 (26.75)	22.12** (8.73)	17.91 (11.40)	23.59** (11.89)
Profitability	-110.31*** (42.47)	-122.42*** (43.45)	-43.73*** (13.58)	-30.44* (16.50)	-35.52** (16.94)
Tangibility	-30.57 (21.26)	-18.41 (23.54)	-11.93* (6.73)	-13.92 (8.76)	-14.13 (9.08)
Current ratio	-4.72** (2.12)	-5.49** (2.23)	-2.74*** (0.73)	-3.80*** (1.22)	-4.16*** (1.24)
Market-to-book	0.12 (2.25)	0.52 (2.36)	-1.14 (0.95)	-1.14 (1.17)	-0.90 (1.19)
Obs.	1,669	1,184	13,158	6,064	5,731
R ²	0.43	0.44	0.57	0.58	0.58

	Panel B			
	(1)	(2)	(3)	(4)
Const.	902.95*** (22.79)	911.75*** (23.14)	909.15*** (23.18)	909.43*** (23.15)
REL(Dummy)	-1.83 (3.45)	-9.74*** (3.11)	-10.92*** (2.44)	-8.44*** (2.88)
REL(Dummy)*Collateral	-15.28*** (4.73)			
REL(Dummy)*Covenant Index		-0.85 (1.68)		
Market-to-Book*Covenant Index			-3.00*** (0.81)	-2.22** (0.95)
REL(Dummy)*Market-to-book*Covenant index				-1.16 (0.75)

(continued overleaf)

Table 7, part 1: Relationships as monitoring commitments.

6.6 Table 7: Relationships and monitoring commitments

Read:

- In single-lender loans, relationship lending still lowers spreads, which weakens the lock-in explanation.
- The relationship benefit is smaller when syndicate size is larger, consistent with lower syndicate moral hazard.
- The relationship benefit is stronger when lead-bank allocation is larger and when loan HHI is higher.
- In panel B, collateral combined with a relationship is associated with lower spreads, suggesting relationships sharpen monitoring incentives.
- Covenant interaction evidence is weaker, but the signs are broadly consistent with the monitoring interpretation.

Economic message: Relationship lending works partly as a commitment device. A prior relationship

reassures syndicate participants that the lead bank has information and incentives to monitor.

6.7 Tables 8 and 9: Collateral and maturity

Read Table 8:

- Relationship measures are negatively related to the probability of pledging collateral.
- $REL(Dummy)$, $REL(Number)$, and $REL(Amount)$ all enter negatively and significantly.
- Larger loan concentration increases the likelihood of collateral.
- The result supports both adverse-selection and monitoring interpretations of collateral.

Read Table 9:

- Relationship loans have shorter maturities on average.
- The maturity effect is nonmonotonic across borrower quality.
- Relationship maturity reductions are strongest for low-quality and high-quality borrowers; middle-quality borrowers show weaker effects.
- The pattern is broadly consistent with Diamond's debt-maturity model.

Economic message: Relationship lending changes the whole contract, not only the spread. Relationships reduce collateral requirements and affect maturity in a way that depends on borrower quality and monitoring needs.

Table 7
(Continued)

Panel B				
	(1)	(2)	(3)	(4)
Covenant index		7.18*** (1.87)	11.22*** (1.90)	11.29*** (1.91)
Log(Maturity)	-18.21*** (2.55)	-18.91*** (2.53)	-18.63*** (2.52)	-18.64*** (2.52)
Collateral	70.72*** (4.72)	49.12*** (3.65)	49.13*** (3.65)	49.16*** (3.64)
Log(Assets)	-9.58*** (1.56)	-9.45*** (1.58)	-9.81*** (1.57)	-9.77*** (1.57)
Log(Loan size)	-16.22*** (1.55)	-16.69*** (1.57)	-16.44*** (1.57)	-16.47*** (1.57)
Log(1+Coverage)	-23.79*** (1.69)	-24.30*** (1.72)	-24.18*** (1.73)	-24.20*** (1.73)
Leverage	22.18** (8.73)	17.78** (8.76)	18.66** (8.74)	18.87** (8.76)
Profitability	-43.74*** (13.64)	-44.63*** (13.64)	-44.83*** (13.64)	-44.69*** (13.63)
Tangibility	-12.07* (6.71)	-10.27 (6.68)	-10.54 (6.68)	-10.74 (6.67)
Current ratio	-2.66*** (0.73)	-2.52*** (0.73)	-2.58*** (0.75)	-2.53*** (0.75)
Market-to-book	-1.11 (0.95)	-0.98 (0.98)	1.59 (1.51)	1.52 (1.49)
Obs.	13,158	13,158	13,158	13,158
R ²	0.57	0.57	0.58	0.58

This table provides the OLS regression (corrected for heteroscedasticity and clustering) estimates of the following equation:

$$AISD = \beta_0 + \beta_1(REL(M)) + \beta_2(Bank\ Monitoring\ Incentive) + \beta_3(REL(M)) \times (Bank\ Monitoring\ Incentive) + \sum \beta_i(Loan_Characteristics_i) + \sum \beta_j(Borrower_Characteristics_j) + \sum \beta_k(Controls_k).$$

The dependent variable AISD is the coupon spread on the drawn amount plus the annual fee in bps. *REL(Dummy)* equals 1 if there is a relationship with any of the lead banks in the last five years before the present loan and 0 otherwise. Maturity is the length in months between facility activation date and maturity date. Loan amount is the loan facility size in millions of real year 2000 dollars. Collateral is a dummy variable that equals 1 if the facility has the stated attribute and 0 otherwise. Log(Assets) is the natural log of the book value of assets of the borrower in millions of year 2000 real dollars as reported in Compustat. Leverage is the ratio of book value of total debt to book value of assets. Coverage is the ratio of EBITDA to interest expenses. Profitability is the ratio of EBITDA to sales. Tangibility is the ratio of NPPE to total assets. Current ratio is the ratio of current assets to current liabilities. Market-to-book is the ratio of (book value of assets – book value of equity + market value of equity) to book value of assets. Syndicate moral hazard, i.e., the incentive to monitor the borrower, is measured by three different proxies: number of members in the syndicate, lead bank allocation of the loan in percent, and the loan concentration in the syndicate measured by the Herfindahl-Hirschmann index following Sufi (2007). Covenant index assumes the value between 0 and 5 with the presence of each of five different covenants coded as 1 and 0 otherwise and summed up (Bradley and Roberts 2004). In addition to the variables reported, the regression also includes industry dummies based on the one-digit SIC code of the borrower, dummies for stated purpose of the facility, dummies for loan type, calendar year dummies, and dummies for the credit rating of the

Table 7, part 2: Covenant and collateral interactions.

Table 8
Lending relationships and probability of pledging collateral

	(1)	(2)	(3)
Const.	-18.15*** (0.62)	-18.15*** (0.63)	-18.16*** (0.63)
<i>REL(Dummy)</i>	-0.22*** (0.05)		
<i>Rel(Number)</i>		-0.26*** (0.06)	
<i>Rel(Amount)</i>			-0.25*** (0.06)
Log(Loan size)	-0.47*** (0.03)	-0.47*** (0.03)	-0.47*** (0.03)
Leverage	2.93*** (0.27)	2.96*** (0.27)	2.96*** (0.27)
Tangibility	-0.48*** (0.16)	-0.48*** (0.16)	-0.48*** (0.16)
Market-to-book	-0.12*** (0.03)	-0.12*** (0.03)	-0.12*** (0.03)
Loan concentration	1.54*** (0.16)	1.57*** (0.16)	1.57*** (0.16)
Log(Maturity)	0.25*** (0.03)	0.26*** (0.03)	0.25*** (0.03)
Obs.	15,510	15,510	15,510
Pseudo R ²	0.26	0.26	0.26

This table provides the logit regression estimates (corrected for heteroscedasticity and clustering) of the following equation:

$$COLLATERAL = \beta_0 + \beta_1(REL(M)) + \beta_2(Log(Loan\ Amount)) + \beta_3(Leverage) + \beta_4(Tangibility) + \beta_5(Market\ to\ Book) + \beta_6(Loan\ Concentration) + \beta_7(Log(Maturity)) + \beta_8(Not\ Rated) + \sum \beta_k(Controls_k).$$

The dependent variable COLLATERAL is a dummy variable that equals 1 if a loan facility is secured by collateral and 0 otherwise. *REL(M)* is the measure of relationship strength, estimated in three different ways: *REL(Dummy)* (equals 1 if there is a relationship with any of the lead banks in the last five years before the present loan and 0 otherwise), *REL(Number)* (ratio of number of deals with the lead bank(s) to total number of loans borrowed by the firm in the last five years before the current loan), and *REL(Amount)* (ratio of the dollar value of deals with the lead bank(s) to the total dollar value of loans borrowed by the firm in the last five years before the current loan). For a facility with multiple lead banks, the maximum *REL(M)* value among all the lead banks is used. The loan amount is the loan facility size in millions of real year 2000 dollars. Leverage is the ratio of book value of total debt to book value of assets. Tangibility is the ratio of NPPE to total assets. Market-to-book is the ratio of (book value of assets – book value of equity + market value of equity) to book value of assets. *Loan concentration* is the ratio of that loan facility amount to sum existing debt and the amount of loan facility. In addition to the variables reported, the regression also includes industry dummies based on the one-digit SIC code of the borrower, dummies for stated purpose of the facility, calendar year dummies, and dummies for the credit rating of the borrower, with not-rated firms considered a separate group. Numbers in parentheses are standard errors corrected for heteroscedasticity and firm-level clustering. Clustering at the deal level produces similar results. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

of borrowing from a relationship lender. The predicted probability from this estimation is used as an instrument for the second-stage probit that estimates the effect of relationship on collateral. We find that the coefficient of the relationship variable for the IV estimation is -1.11 and it is significant at the 1% level (for comparison, the coefficient on relationships in a probit equation (not correcting for endogeneity) for collateral generates a coefficient of -0.13, significant at the 1% level). Note that the results reported in Table 8 use a logit specification where the coefficient for the relationship variable is -0.22,

Table 8: Lending relationships and probability of pledging collateral.

Table 9
Lending relationships and loan maturity

	(1)	(2)	(3)	(4)	(5)	(6)
Const.	2.58*** (0.10)	2.65*** (0.10)	3.42*** (0.23)	3.73*** (0.23)	2.57*** (0.13)	2.54*** (0.10)
<i>REL</i> (<i>Dummy</i>)	-0.08*** (0.02)		-0.09*** (0.04)	-0.05* (0.03)	-0.09*** (0.02)	-0.08*** (0.02)
<i>REL</i> (<i>Dummy</i>) × High		-0.10*** (0.02)				
<i>REL</i> (<i>Dummy</i>) × Middle		-0.01 (0.02)				
<i>REL</i> (<i>Dummy</i>) × Low		-0.15*** (0.03)				
Log(Loan size)	0.14*** (0.009)	0.13*** (0.009)	0.09*** (0.02)	0.13*** (0.02)	0.14*** (0.01)	0.14*** (0.009)
Leverage	0.41*** (0.04)	0.38*** (0.04)	0.36*** (0.11)	0.38*** (0.08)	0.39*** (0.05)	0.41*** (0.04)
Log(Assets)	-0.05*** (0.009)	-0.05*** (0.009)	-0.03* (0.02)	-0.08*** (0.02)	-0.03*** (0.01)	-0.05*** (0.009)
Market-to-book	-0.005 (0.005)	-0.005 (0.005)	0.006 (0.01)	0.02 (0.02)	-0.008 (0.006)	-0.005 (0.005)
Log (Asset maturity)	0.03*** (0.01)	0.04*** (0.01)	0.03 (0.02)	-0.04 (0.02)	0.06*** (0.01)	0.03*** (0.01)
Regulated	-0.47*** (0.06)	-0.47*** (0.05)	-0.30*** (0.06)	-0.52*** (0.13)	-0.49*** (0.10)	-0.47*** (0.06)
Collateral	0.12*** (0.02)	0.12*** (0.02)	0.26*** (0.06)	0.14*** (0.03)	0.08*** (0.02)	0.12*** (0.02)
Term spread						0.0004** (0.0002)
Obs.	15,636	15,636	3,174	3,289	9,173	15,636
<i>R</i> ²	0.25	0.25	0.33	0.27	0.21	0.25

This table provides the OLS regression (corrected for heteroscedasticity and clustering) estimates of the following equation:

$$\begin{aligned} \text{Log}(\text{Maturity}) = & \beta_0 + \beta_1(\text{REL}(M)) + \beta_2(\text{Log}(\text{Loan Amount})) + \beta_3(\text{Leverage}) + \beta_4(\text{Log}(\text{Assets})) \\ & + \beta_5(\text{Market to Book}) + \beta_6(\text{Log}(\text{Asset Maturity})) + \beta_7(\text{Regulated}) + \beta_8(\text{Collateral}) \\ & + \sum \beta_k(\text{Control}_k). \end{aligned}$$

The dependent variable Log(Maturity) is the natural log of the stated maturity of the loan facility (measured as the length in months between the facility activation date and maturity date). *REL*(*Dummy*) equals 1 if there is a relationship with any of the lead banks in the last five years before the present loan and 0 otherwise. Log(assets) is the natural log of the book value of assets of the borrower. Loan amount is the loan facility size. Leverage is the ratio of book value of total debt to book value of assets. Market-to-book is the ratio of (book value of assets – book value of equity + market value of equity) to book value of assets. Asset maturity is the weighted average of current assets divided by the cost of goods sold, and Net PPE divided by depreciation and amortization—as defined in Barclay, Marx, and Smith (2003). *Regulated Industry* is a dummy variable that equals 1 for firms in the utilities industry under the Fama-French (1997) industry classification and 0 otherwise. Term spread is the difference in yields between one-year and ten-year U.S. government bonds at the time of loan activation. In addition to the variables reported, the regression also includes industry dummies based on the one-digit SIC code of the borrower, dummies for the stated purpose of the facility, and calendar year dummies. Numbers in parentheses are standard errors corrected for heteroscedasticity and firm-level clustering. Clustering at the deal level produces similar results. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

lenders commit to monitoring the firms more with shorter maturity loans. Given that monitoring costs for relationship lenders are likely to be lower, we expect relationship loans to have a shorter maturity for low-quality borrowers. These results are consistent with Diamond (1991). The last column reports the association between maturity and the prevailing term spread. We find a significant

Table 10
Instrumental variables estimation of loan spread, maturity, and collateral

Dependent variable	AISD			Log(Maturity)		Collateral	
	OLS	IV	IV	OLS	IV	Probit	IV Probit
<i>REL</i> (<i>Dummy</i>)	-10.55*** (2.44)	-14.35** (5.73)	-12.79** (5.73)			-0.13*** (0.03)	-0.07*** (0.31)
<i>REL</i> (<i>Dummy</i>) × High				-0.10*** (0.02)	-0.02 (0.03)		
<i>REL</i> (<i>Dummy</i>) × Middle				-0.01 (0.02)	0.07*** (0.03)		
<i>REL</i> (<i>Dummy</i>) × Low				-0.15*** (0.03)	-0.10*** (0.04)		
Log(Maturity)	-18.07*** (2.54)	-151.6*** (42.70)	-146.13*** (42.04)			0.18*** (0.02)	0.94*** (0.15)
Collateral	60.01*** (2.87)	406.2*** (68.09)	408.2*** (64.94)	0.12*** (0.02)	1.67*** (0.97)		
Default spread							
Average spread			0.68*** (0.11)				
Log(Asset maturity)				0.04*** (0.01)	0.65*** (0.15)		
Regulated				-0.47*** (0.05)	-0.24*** (0.77)		
Loan concentration						0.90*** (0.09)	1.00*** (0.07)
Other controls	As in column 1 of Table 4	As in column 1 of Table 4	As in column 1 of Table 4	As in column 2 of Table 9		As in column of Table 8	
Tests of endogeneity: Durbin-Wu-Hausman's chi-square test	469.25	546.46		1170			
p-value	0.00	0.00		0.00			
Wald's chi-square test statistic						29.56	
p-value						0.00	

Table 9: Lending relationships and loan maturity.

Table 10, part 1: Joint IV estimation of spread, maturity, and collateral.

6.8 Tables 10 and 11: Joint contract determination and credit availability

Read Table 10:

- The paper estimates spread, collateral, and maturity jointly using IV methods.
- Relationship lending continues to reduce spreads, with IV coefficients around -12.8 to -14.4 bps in the spread equation.
- Relationship lending reduces the probability of collateral in the IV probit specification.
- Instrument relevance and validity tests generally support the IV design.

Read Table 11:

- Relationship loans are larger when scaled by borrower assets.
- Relationship loans are also larger when scaled by the borrower's long-term debt.
- The positive effect appears for all three relationship measures.

- This suggests that relationships improve credit availability, not just loan pricing.

Economic message: The central findings survive a system that allows contract terms to be jointly determined. Relationships lower borrowing costs, lower collateral requirements, and expand access to bank debt.

Dependent variable	AISD	Log(Maturity)	Collateral
<i>Weak identification statistics:</i>			
Cragg-Donald F -stat	14.57	15.05	
Stock and Yogo (2005) critical value	11.03	11.03	
First-stage F -statistic			323.27
Partial R -square		8.16%	
Anderson's LR statistic	58.24	60.17	1289
p -value	0.00	0.00	0.00
<i>Instrument exogeneity test:</i>			
Hansen's J -statistic	4.132	3.967	0.001
p -value	0.13	0.14	0.98

This table provides estimations of the following system of equations using instrumental variables to estimate the impact of past relationships on Loan Spread, Maturity, and Collateral:

$$\begin{aligned} AISD &= \gamma_1(REL) + \gamma_{1C}(Collateral) + \gamma_{1M}(Log(Maturity)) + X_A\beta_A + \epsilon_A \\ Collateral &= \gamma_2(REL) + \gamma_{2C}(Log(Maturity)) + X_C\beta_C + \epsilon_C \\ Log(Maturity) &= \gamma_3(REL) + \gamma_{3C}(Collateral) + X_M\beta_M + \epsilon_M \end{aligned}$$

AISD is the coupon spread over LIBOR on the drawn amount plus the annual fee. Log(Maturity) is the natural log of the stated maturity of the loan facility (measured as the length in month between the facility activation date and maturity date). Collateral is a dummy variable that equals 1 if the loan was secured and 0 otherwise. $REL(Dummy)$ is 1 if there is a relationship with any of the lead banks in the last five years before the present loan and 0 otherwise. Default spread is measured as the difference between the yields on Moody's seasoned corporate bond with Baa rating and ten-year U.S. government bonds. Asset maturity is the weighted average of current assets divided by the cost of goods sold, and Net PPE divided by depreciation at amortization—as defined in Barclay, Marx, and Smith (2003). *Repledged industry* is a dummy variable that equals 1 for firms in the utilities industry under the Fama-French (1997) industry classification and 0 otherwise. *Loan concentration* is the ratio of that loan facility amount to sum existing debt and the amount of loan facility. The Durbin-Wu-Hausman chi-square test the test of the null hypothesis that maturity and collateral are indeed exogenous. The rejection of null implies they are endogenous. Wald's test provides a similar test for IV probit model for collateral. To test for instrument relevance, for AISD regression since we have two endogenous regressors, we estimate the Cragg-Donald statistic and compare it against critical value in Table 1 of Stock and Yogo (2005). For the Log(Maturity) regression, we report the first-stage F -statistic, as well as partial R^2 since it has only one endogenous variable. For the AISD and Log(Maturity) IV regressions, we report the Anderson LR test of the null hypothesis that our instruments and endogenous variables are not correlated. We reject the null (p -value = 0.00), implying that our instruments are correlated with the endogenous variables. Since we have two endogenous variables but four instruments for the spread equation, we are able to estimate the Hansen J -statistic for overidentification restrictions. These are just tests of the null hypothesis that the correct model is specified and the orthogonality condition are met (correlations between the instruments and the error term is zero). Our test statistic does not reject the null for AISD and Log(Maturity) regressions implying exogeneity of if instruments. For both AISD and Log(Maturity) IV regressions, numbers in the parentheses are standard errors corrected for heteroscedasticity and firm-level clustering. For IV robust

Table 10, part 2: Weak-instrument and overidentification diagnostics.

6.9 Appendix tables: Sensitivity to hidden bias

Read:

- Table A1 reports Rosenbaum bounds for the propensity-score matching estimates.
- The relationship-spread effect remains negative until hidden bias becomes large enough to raise the treatment odds ratio to about 1.5.
- Table A2 translates this hidden-bias magnitude into movements in observed covariates.
- For example, leverage would need to move by about 2.73 standard deviations to generate the same odds-ratio shift.

Table 11
Lending relationships and access to loans

	(1)	(2)	(3)	(4)	(5)	(6)
Const.	0.63*** (0.03)	0.63*** (0.03)	0.63*** (0.03)	1.43*** (0.06)	1.42*** (0.06)	1.42*** (0.06)
$REL(Dummy)$	0.008*** (0.004)			0.02*** (0.007)		
$REL(Number)$		0.009** (0.004)			0.03*** (0.007)	
$REL(Amount)$			0.01*** (0.004)			0.03*** (0.007)
Log(Assets)	-0.06*** (0.002)	-0.06*** (0.002)	-0.06*** (0.002)	-0.10*** (0.003)	-0.10*** (0.003)	-0.10*** (0.003)
Log(1+Coverage)	0.001 (0.002)	0.001 (0.002)	0.001 (0.002)	0.13*** (0.005)	0.13*** (0.005)	0.13*** (0.005)
Log(Syndicate size)	0.06*** (0.002)	0.06*** (0.002)	0.06*** (0.002)	0.07*** (0.004)	0.07*** (0.004)	0.07*** (0.004)
Profitability	0.13*** (0.02)	0.13*** (0.02)	0.13*** (0.02)	-0.34*** (0.04)	-0.34*** (0.04)	-0.34*** (0.04)
Tangibility	-0.02** (0.01)	-0.02** (0.01)	-0.02** (0.01)	-0.15*** (0.02)	-0.15*** (0.02)	-0.15*** (0.02)
Current ratio	0.0002 (0.001)	0.0002 (0.001)	0.0002 (0.001)	-0.004 (0.002)	-0.004 (0.002)	-0.004 (0.002)
Market-to-book	0.01*** (0.003)	0.01*** (0.003)	0.01*** (0.003)	0.008** (0.003)	0.008** (0.003)	0.008** (0.003)
Obs.	15,218	15,218	15,218	14,626	14,626	14,626
R^2	0.32	0.32	0.32	0.42	0.42	0.42

The first three specifications of this table provide the OLS estimates (corrected for heteroscedasticity) of the following equation:

$$\begin{aligned} \frac{Loan\ Amount}{Assets} &= \beta_0 + \beta_1(REL(M)) + \sum \beta_i(Loan_Characteristics_i) \\ &+ \sum \beta_j(Borrower_Characteristics_j) + \sum \beta_k(Controls_k). \end{aligned}$$

The next three specifications are essentially the same but the dependent variable is $\frac{Loan\ Amount}{Loan\ Term\ Book}$, $REL(M)$ is the measure of relationship strength, estimated in three different ways: $REL(Dummy)$ equals 1 if there is a relationship with any of the lead banks in the last five years before the present loan and 0 otherwise, $REL(Number)$ (the ratio of number of deals with the lead bank(s) to the total number of loans borrowed by the firm in the last five years before the current loan), and $REL(Amount)$ (the ratio of dollar value of deals with the lead bank(s) to the total dollar value of loans borrowed by the firm in the last five years before the current loan). For a facility with multiple lead banks, the maximum $REL(M)$ value among all the lead banks is used. The loan amount is the loan facility size in millions of real year 2000 dollars. Profitability is the ratio of EBITDA to sales. Tangibility is the ratio of PPE to total assets. Current ratio is the ratio of current assets to current liabilities. Market-to-book is the ratio of (book value of assets – book value of equity + market value of equity) to book value of assets. In addition to the variables reported, the regression also includes industry dummies based on the one-digit SIC code of the borrower, dummies for stated purpose of the facility, calendar year dummies, and dummies for the credit rating of the borrower, with not-rated firms considered a separate group. Numbers in parentheses are standard errors corrected for heteroscedasticity and firm-level clustering. Clustering at the deal level produces similar results. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

and transactional lending. We find that spreads charged for relationship loans and nonrelationship loans become indistinguishable if the borrower was in the top 30% when ranked by the asset size. Similar dissipation of relationship benefits occurs if the borrower has a rated public debt or is part of the S&P 500 index. The article uses the loan syndicate structure to examine the interaction of past relationships and syndicate moral hazard due to multiple lenders. The results in

Table 11: Lending relationships and access to loans.

Economic message: Selection on unobservables would need to be quite strong to overturn the matching evidence that relationship lending lowers spreads.

Rosenbaum bounds

exp(γ)	p -critical		Effect on spreads	
	Upper bound	Lower bound	Upper bound	Lower bound
1	0	0	-22.88	-19.06
1.05	0	0	-25.25	-16.63
1.1	0	0	-27.50	-14.29
1.15	0	0	-29.63	-12.05
1.2	0	0	-31.69	-9.88
1.25	0	6.70E-16	-33.63	-7.81
1.3	0	1.20E-10	-35.50	-5.79
1.35	0	9.80E-07	-37.25	-3.88
1.4	0	0.000609	-38.94	-2.00
1.45	0	0.038583	-40.56	-0.13
1.50	0	0.363088	-42.13	1.63

Rosenbaum (2002) shows that for a given value of γ , the null distribution of $T = t(Z, r)$ is bounded by two known distributions for T^+ and T^- , where

$$F(T^+) = \sum_{d=1}^S d \cdot n^+.$$

Table A1: Rosenbaum bounds for hidden bias.

Table A2
Examination of hidden bias n standard deviation in observed covariate required to justify an increase in odds ratio by a factor of 1.5 holding all other variables constant in the propensity score model

Variable	Odds ratio exp(β_k)	Std. dev. s_k	Implied n
Assets	0.987	1.872	-16.55
Profitability	0.899	0.120	-31.73
Log(1+Coverage)	1.189	1.106	2.12
Leverage	2.194	0.189	2.73
Default spread	1.117	0.582	6.30

Although such unmeasured influences must have in order that the estimated treatment effects from propensity score matching would have arisen purely through nonrandom assignment. Intuitive interpretation of exp(γ) being equal to 1 is that unobservable factor(s) (hidden bias) has no effect on each individual record (in our case each loan facility) is assigned to a treatment group, being classified as a relationship loan) or to a control group (e.g., being classified as a relationship loan). For values of exp(γ) higher than 1, the implication is that the hidden bias is

Table A2: Observed-covariate changes needed to justify large hidden bias.

7 Short summary

- The paper studies whether repeated borrowing from the same lender improves loan contract terms for public firms.
- Relationship lending is defined using prior lead-lender relationships during the previous five years.
- Relationship loans have lower raw spreads and better nonprice terms than nonrelationship loans.
- After detailed loan and borrower controls, relationship loans still carry spreads around 10–11 bps lower.
- Matching, IV, and treatment-effects approaches all support the conclusion that relationships lower spreads.
- Relationship benefits are concentrated among opaque borrowers: smaller firms, unrated firms, firms outside the S&P 500, and firms with fewer analysts.
- The paper identifies a boundary where relationship lending becomes indistinguishable from transactional lending.
- Relationships mitigate syndicate moral hazard by making lead-bank monitoring more credible.
- Relationship loans are less likely to require collateral.
- Relationship effects on maturity depend on borrower quality and are consistent with monitoring theories.
- Relationship borrowers obtain larger loans relative to assets and long-term debt.
- The overall message is that private information and monitoring relationships still matter in modern syndicated lending.

8 Conclusion

8.1 Core conclusion

Bharath, Dahiya, Saunders, and Srinivasan show that lending relationships improve loan contract terms even for publicly traded firms. Borrowing from a prior lead lender is associated with lower spreads, lower collateral requirements, and larger loan availability.

8.2 Economic intuition

A relationship lender has accumulated borrower-specific information that outside lenders do not have. This lowers screening and monitoring costs. In syndicated lending, the relationship also signals that the lead bank has both the information and the incentive to monitor. These informational and monitoring advantages are passed partly to borrowers through better loan terms.

8.3 Incremental contribution revisited

The paper's most important contribution is to separate relationship lending from transactional lending. Relationship benefits are not universal. They are strongest when the borrower is opaque and disappear when public information is abundant. This provides a sharper empirical interpretation of what relationship lending is economically doing.

8.4 Implications

The findings imply that bank relationships remain valuable even in deep capital markets. Public firms may still prefer prior lenders because relationships lower contracting frictions. For banks, the results imply that relationship capital can create pricing and monitoring advantages, especially with opaque borrowers and complex syndicated loans.

8.5 Limitations

The paper relies on nonrandom relationship formation, so causal interpretation depends on the strength of the matching, IV, and treatment-effects assumptions. The distance instrument is plausible but not directly testable. The paper also cannot fully observe all cross-product relationship benefits, such as underwriting, cash management, or other banking services.

8.6 Takeaway

The enduring takeaway is that relationship banking creates measurable value by reducing information and monitoring frictions. In public-firm syndicated loans, relationship lenders offer lower spreads, require less collateral, support larger borrowing, and help distinguish relationship lending from transactional lending.